

Interactive Data Visualization: Open Data of Pandemic Disease

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Abstract— A pandemic disease or pandemic disease is an infectious disease with a wide area of distribution, spread in several countries or even throughout the world. Several types of pandemic diseases that have existed include Severe Acute Respiratory Syndrome and Ebola. The development of the pandemic disease is one of the concerns of the world health organization. Open data (open data) on the spread of pandemic diseases is published to make it easier for related parties such as the government, health organizations to the public to obtain an overview of the problem of pandemic disease. However, published open data is still presented in numerical form so that the expected objectives are not achieved. In this research, data visualization will be carried out using open data on pandemic diseases. The research stages to be carried out were preceded by data analysis. The next stage is the clustering process, and the last stage is interactive data visualization. Literature studies were carried out during this research. The results in this study are interactive data visualization of pandemic disease open data, clustered by country and type of pandemic disease, cumulative number of cases, cumulative number of deaths and pandemic disease intersection in each country.

Keywords—component, formatting, style, styling, insert

I. INTRODUCTION

A pandemic disease or pandemic disease is an infectious disease with a wide area of distribution, spread in several countries or even throughout the world. Several types of pandemic diseases that have existed include Severe Acute Respiratory Syndrome (SARS) and Ebola. SARS was first identified in China (currently known as the People's Republic of China) in November 2002, and spread to 26 other countries [1]–[5]. The first Ebola outbreak occurred in remote villages in Central Africa, until in 2014 - 2016 this outbreak was the largest outbreak in West Africa and spread between countries [2]. The pandemic disease that is currently being faced by the whole world is the Coronavirus Disease (COVID-19). Currently, there is no vaccine or specific treatment for COVID-19. However, there are many ongoing clinical trials to evaluate the potential treatments that can be performed [6], [7].

The development of the pandemic disease is one of the concerns of the world health organization. Open data (open data) on the spread of pandemic diseases is published to make it easier for related parties such as the government, health organizations to the public to obtain an overview of the problem of pandemic disease. However, published open data is still presented in numerical form (table) so that the expected objectives are not achieved. On the other hand, the development of research in the field of data continues to develop, open data is also used by data scientists to be processed to produce data visualizations. Data visualization

is a graphical representation of data to help users understand the context and significance of that data. Interactive data visualizations allow users to drill down into data details, identify patterns and outliers and change which data is processed and / or excluded. While data visualization is a technique used to communicate data by making it into visual objects [8]–[14].

Previous research, regarding the visualization of population data, revealed that with a data model that was depicted in a visual form, it made it easier for the village government to see trends in the village population and assist in exploring the potential of the village [15], [16]. Other research reveals that interactive data visualization makes it easier for users to read open data information from the DKI Jakarta Provincial Government on the topic of regional economy and finance [17].

In this research, data visualization will be carried out using open data on pandemic diseases. The expected results in this study are interactive data visualization of pandemic disease open data, clustered by country and type of pandemic disease, cumulative number of cases, cumulative number of deaths and pandemic disease intersection in each country.

II. RESEARCH METHOD

As has been stated by Ben Fry, the success of communication through data visualization is determined by how the data is visualized so that communication opponents understand it. There are 7 data visualization processes namely [18]–[22]:

- 1 Acquire; get data, either from files on disk or over a network
- 2 Parse; provide some structure for the meaning of the data and organize them into categories.
- 3 Filter; delete unnecessary data.
- 4 Mine; apply statistical or data mining methods to see patterns or place data in a mathematical context.
- 5 Represent; choose a basic visual model, such as a bar, list or tree chart.
- 6 Refine; enhance the basic representation to make it clearer and more visually appealing.
- 7 Interact; add methods to manipulate data or control viewable features.

Pandemic disease open data used in this study were Severe Acute Respiratory Syndrome (SARS) and Ebola Virus Disease outbreaks. Data sources were obtained from WHO pandemic report cases. The time period of the data used is as follows:

- 1 Severe Acute Respiratory Syndrome (SARS): March - July 2003
- 2 Ebola Virus Disease: August 2014 - March 2016

Based on the data used, this study does not discuss data related to new cases and new deaths from pandemic diseases. The number of open data records for Severe Acute Respiratory Syndrome (SARS) was 2,538 records and Ebola Virus Disease was 2,485 records.

Based on the open data obtained from pandemic diseases, the first stage of this research is the data analysis stage, the process of exploring open pandemic disease data is carried out in the database, analyzed (tables of queries) and generating data queries. This stage is used to answer the information needs of each data which is then mapped to the clustering method (grouping).

Furthermore, based on the data query, clustering is carried out using the clustering method. Clustering is carried out to minimize data diversity in a data group and maximize data diversity between data groups. The results of the clustering stage are country data and country pandemic disease, number of cases of pandemic disease, number of deaths from pandemic disease, number recovered and intersection of pandemic disease in each country. The last stage is interactive data visualization using visualization tools on clustered data. The research chart is illustrated in Fig 1.

The result of this research is interactive data visualization of pandemic disease open data, clustered by country and type of pandemic disease, cumulative number of cases, cumulative number of deaths, cumulative number of recovered, and pandemic disease intersection in each country.

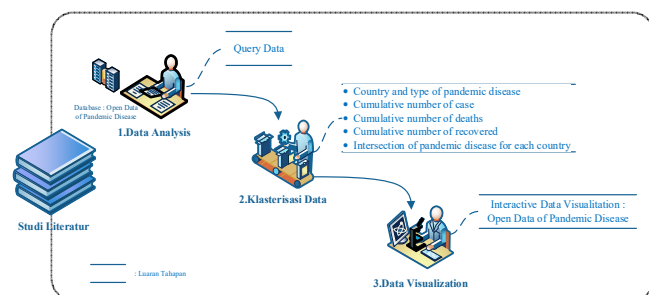


Fig 1. Research Model

III. RESULT AND DISCUSSION

A. Pandemic Data Analysis

Based on the data obtained, there are a total of 5023 pandemic report data records consisting of the Severe Acute Respiratory Syndrome (SARS) and Ebola Virus Disease pandemic report. In the initial stage, 5023 data records were parsed, by providing a data structure and the data were organized into categories, namely:

1. State and pandemic disease that occurred
2. Number of cases / countries
3. Number of deaths / countries

4. The number of recoveries / countries
5. Pandemic disease intersection for each country

The next step is to filter by removing unnecessary data and obtain 5 fields that will be used in data visualization, namely Country, Cumulative number of cases, Number of deaths, Number recovered and Pandemic Disease. Each field has a data type and data range as in Fig 2.

Name	Type	Range	Missings
Country	polynomial	=[Australia, Belgium, Brazil, ...]	= 0
Cumulative number of case...	integer	=[0 - 8704]	= 1
Number of deaths	integer	=[0 - 3858]	= 837
Number recovered	integer	=[-711 - 5196]	= 0
Pandemi Disease	polynomial	=[EBOLA, SARS]	= 0

Fig 2. Field Data

There is 1 value in the Cumulative number of cases field and 837 Number of deaths. Data cleaning is performed on the missing value, namely replacing the missing value with a minimum value, namely zero. The results of the cleaning data are as in Fig 3.

Data: SimpleExampleSet: 5023 examples, 5 regular attributes, no special attributes
 expects: ExampleSetMetaData: #examples: = 0; #attributes: 0;
 Example set matching at least one selected attribute.

Name	Type	Range	Missings
Country	polynomial	=[Australia, Belgium, Brazil, Bulgaria, ...]	= 0
Cumulative numbe...	integer	=[0 - 8704]	= 0
Number of deaths	integer	=[0 - 3858]	= 0
Number recovered	integer	=[-711 - 5196]	= 0
Pandemi Disease	polynomial	=[EBOLA, SARS]	= 0

Fig 3. Data Cleaning

The data distribution matrix for Cumulative number of cases, Number of deaths, Number recovered by country is shown in Fig 4.

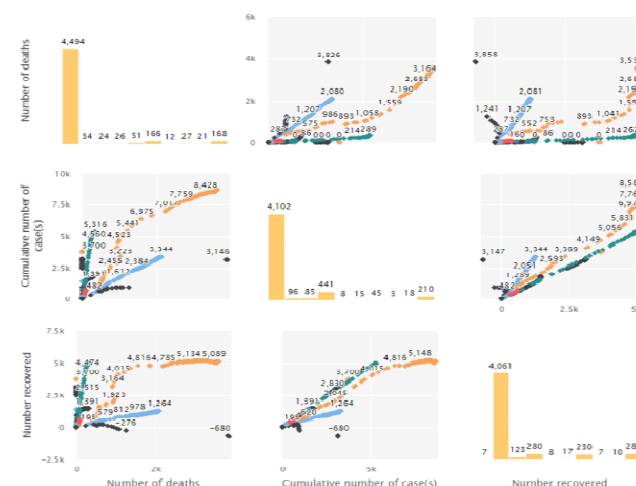


Fig 4. Data Distribution Matrix

B. Pandemic Data Clustering and Visualization

Pandemic data clustering uses the k-means clustering algorithm. This clustering algorithm is widely used because of the simple clustering method to identify the number of

clusters that have been predetermined. The k-means algorithm determines a set of k clusters and assigns each data to an exact cluster. Clusters consist of similar data, and similarities between data are based on a measure of the distance between them. Clusters in the k-means algorithm are determined by the position of the center in the n -dimensional space of the n data attributes called centroids. The k-means algorithm starts with a point k which is treated as the center of the k cluster. All data were assigned to the nearest cluster using euclidian distance measurements. Furthermore, the centroid in each cluster is recalculated by averaging all data in each cluster. The previous step is repeated for obtaining a new centroid, until the centroid is no longer moved, or an optimization step is reached [23]–[28]. In pandemic data, clustering is carried out on 2 of 3 variables with a value of k for each cluster = 2. The clustering in question is:

1. Cluster A uses the variable number of cases and number of deaths.
2. Cluster B uses the variables Cumulative number of cases and Number recovered.
3. Cluster C uses the variables Number of deaths and Number recovered.

1) Clusterization A

Based on the clusterization carried out with the cumulative number of cases and Number of deaths variables, the final centroid value is obtained, which can be seen in Fig. 5.

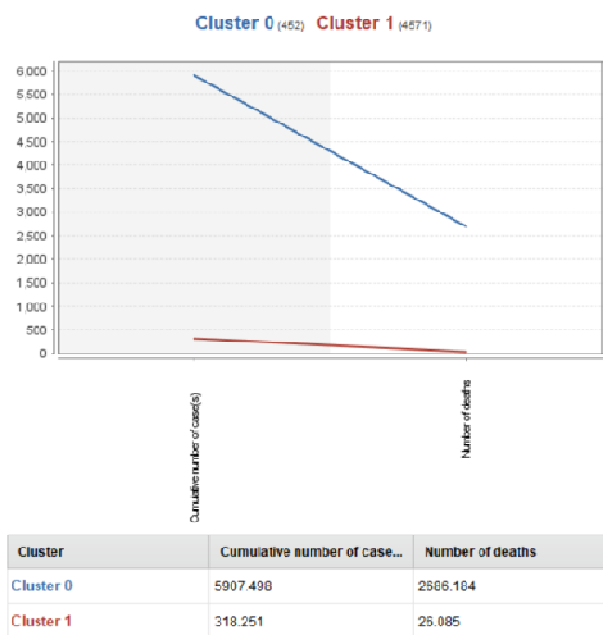


Fig. 5. Final Value of Centroid – Clustering A

The results of clustering produce 2 clusters, namely: Cluster 0 and Cluster 1. Clustering is presented in a scatter plot as in Fig. 6 and Fig. 7. Clusterization is presented in a scatter plot to show the relationship between 2 (expressed in variable X and variable Y). This diagram also determines the type of relationship of the 2 variables whether positive, negative or nonlinear. The resulting plot pattern in the scatter

plot cluster 0 and cluster 1 in cluster A shows a weak degree of correlation and forms a curve, so it indicates a nonlinear relationship in the cluster. The cumulative number of cases variable affects the Number of deaths variable, but the degree of influence is weak or uncertain.

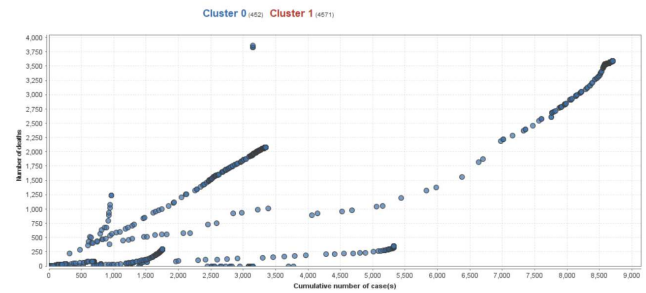


Fig. 6. Scatter Plot Cluster 0 – Clustering A

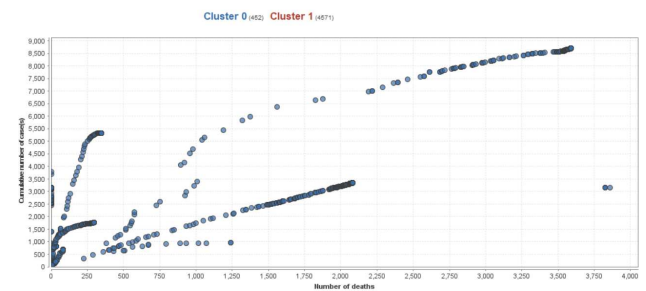


Fig. 7. Scatter Plot Cluster 1 – Clustering A

2) Clusterization B

Based on the clusterization carried out with the cumulative number of cases and Number recovered variables, the final centroid value is obtained, which can be seen in Fig. 8.

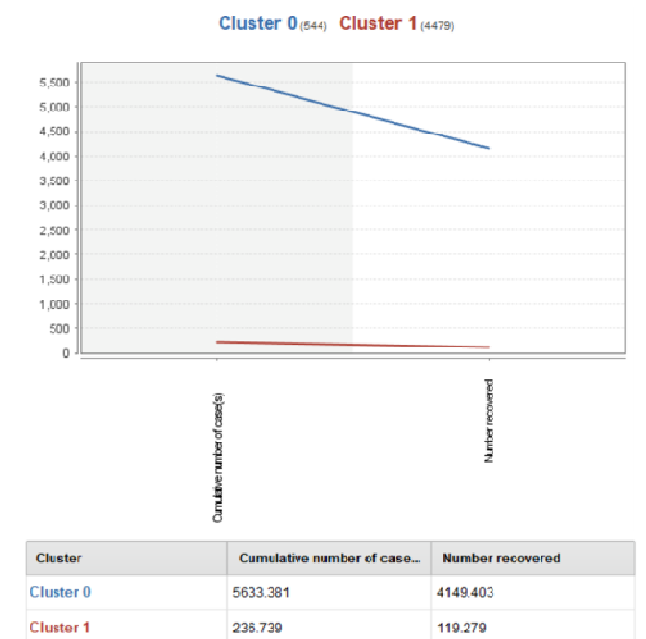


Fig. 8. Final Value of Centroid – Clustering B

The results of clustering produce 2 clusters, namely: Cluster 0 and Cluster 1. Clustering is presented in a scatter plot as in Fig. 9 and Fig. 10. Scatter plot cluster 0 and cluster 1 in cluster B produce the same plot pattern. This pattern also shows the degree of weak correlation and forms a curve, so it indicates a nonlinear relationship in the cluster. The cumulative number of cases variable affects the Number recovered variable, but the level of the effect is weak or uncertain.

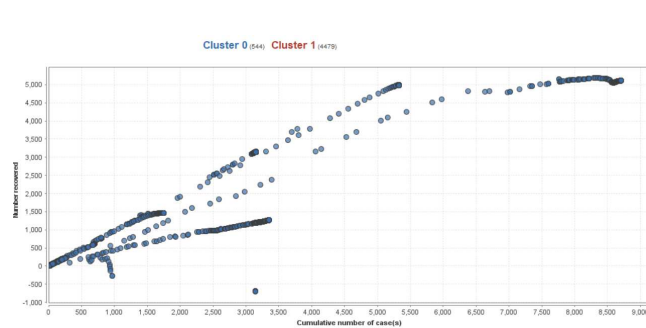


Fig. 9. Scatter Plot Cluster 0 – Clustering B

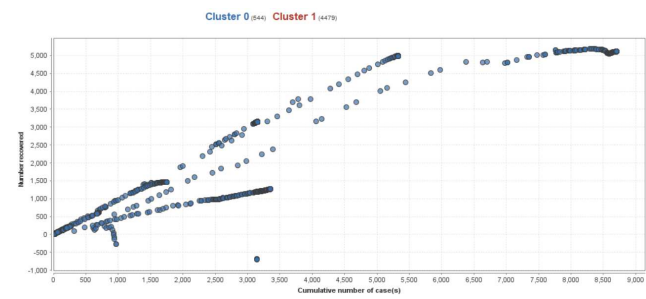


Fig. 10. Scatter Plot Cluster 1 – Clustering B

3) Clusterization C

Based on the clusterization carried out with the Number of deaths and Number recovered variables, the final centroid value is obtained, which can be seen in Fig. 11.



Fig. 11. Final Value of Centroid – Clustering C

The results of clustering produce 2 clusters, namely: Cluster 0 and Cluster 1. Clustering is presented in a scatter plot as in Fig. 12 and Fig. 13. Scatter plot cluster 0 and cluster 1 in cluster C produce the same plot pattern. As with clusterization A and clusterization B, this pattern also shows a weak degree of correlation and forms a curve, so it indicates a nonlinear relationship in the cluster. The variable Number of deaths affects the Number recovered variable, but the level of the effect is weak or uncertain.

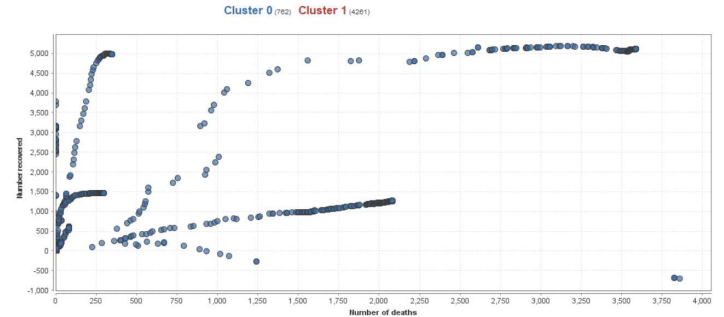


Fig. 12. Scatter Plot Cluster 0 – Clustering C

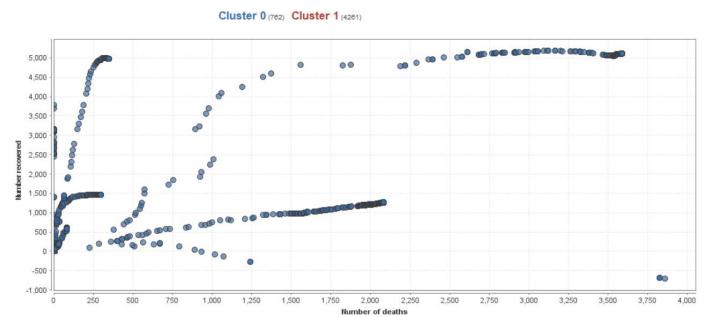


Fig. 13. Scatter Plot Cluster 1 – Clustering C

IV. CONCLUSION

Based on the research conducted, the following conclusions can be drawn:

1. Data analysis has been carried out and determined 5 fields to be used in the data visualization, namely Country, Cumulative number of cases, Number of deaths, Number recovered and Pandemic Disease.
2. The data cleaning stage is performed on the missing value, namely the Cumulative number of cases and Number of deaths field. Data cleaning is done by replacing the missing value with a minimum value of zero.
3. Pandemic data clustering / clustering is carried out using the k-means clustering algorithm.
4. The resulting scatter plot pattern in cluster A, B and C shows the degree of weak correlation and forms a curve, so that it indicates a nonlinear relationship in the cluster.

REFERENCES

- [1] WHO, "Laboratory biosafety manual Third edition World Health Organization," *World Heal. Organ.*, pp. 1–178, 2004.
- [2] D. Malvy, A. K. McElroy, H. de Clerck, S. Günther, and J. van Griensven, "Ebola virus disease," *Lancet*, vol. 393, no. 10174, pp. 936–948, 2019.

- [3] Y. Waheed, "Ebola in West Africa: An international medical emergency," *Asian Pac. J. Trop. Biomed.*, vol. 4, no. 9, pp. 673–674, Sep. 2014.
- [4] H. H. Huang, D. H. T. Yen, W. F. Kao, L. M. Wang, C. I. Huang, and C. H. Lee, "Declining emergency department visits and costs during the severe acute respiratory syndrome (SARS) outbreak," *J. Formos. Med. Assoc.*, vol. 105, no. 1, pp. 31–37, Jan. 2006.
- [5] H. J. Chang, N. Huang, C. H. Lee, Y. J. Hsu, C. J. Hsieh, and Y. J. Chou, "The Impact of the SARS Epidemic on the Utilization of Medical Services: SARS and the Fear of SARS," *Am. J. Public Health*, vol. 94, no. 4, pp. 562–564, 2004.
- [6] WHO, "Coronavirus," 2020. [Online]. Available: https://www.who.int/health-topics/coronavirus#tab=tab_1. [Accessed: 30-Mar-2020].
- [7] Q. Li *et al.*, "Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus-Infected Pneumonia," *N. Engl. J. Med.*, vol. 382, no. 13, pp. 1199–1207, Mar. 2020.
- [8] M. Friendly, C. Chen, W. Härdle, and A. Unwin, "A Brief History of Data Visualization," *Handb. Data Vis.*, pp. 15–56, 2008.
- [9] A. Buja, D. F. Swayne, M. L. Littman, N. Dean, H. Hofmann, and L. Chen, "Data Visualization with Multidimensional Scaling," *J. Comput. Graph. Stat.*, vol. 17, no. 2, pp. 444–472, 2008.
- [10] W. Chen, F. Guo, and F. Y. Wang, "A Survey of Traffic Data Visualization," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 6, pp. 2970–2984, 2015.
- [11] P. Godfrey, J. Gryz, and P. Lasek, "Interactive Visualization of Large Data Sets," *IEEE Trans. Knowl. Data Eng.*, vol. 28, no. 8, pp. 2142–2157, Aug. 2016.
- [12] Lousa, Andre, I. Pedrosa, and J. Bernardino, "Evaluation and Analysis of Business Intelligence Data Visualization Tools," in *2019 14th Iberian Conference on Information Systems and Technologies (CISTI)*, 2019, pp. 1–6.
- [13] Wang, Deqiang, D. Guo, and Z. Hui, "Spatial temporal data visualization in emergency management: a view from data-driven decision," in *Proceedings of the 3rd ACM SIGSPATIAL Workshop on Emergency Management using*, 2017, pp. 1–7.
- [14] C. M. Parish and P. D. Edmondson, "Data visualization heuristics for the physical sciences," *Mater. Des.*, vol. 179, p. 107868, Oct. 2019.
- [15] E. Lestariningsih, E. Ardianto, W. T. Handoko, E. Supriyanto, and S. L. R. A., "Visualisasi Data Penduduk Berbasis Web di Kelurahan Mranggen Kecamatan Mranggen Kabupaten Demak menggunakan Highcart 5.0.6," *J. Teknol. Inf. Din.*, vol. 21, no. 2, pp. 146–153, 2016.
- [16] W. Hidayat and N. Rahardjo, "Visualisasi Data Jumlah Penduduk Dalam Bentuk Kartogram (Studi Kasus Di Daerah Istimewa Yogyakarta)," *J. Bumi Indones.*, vol. 3, no. 2, pp. 1–31, 2013.
- [17] N. A. Syaripul and A. M. Bachtiar, "Visualisasi Data Interaktif Data Terbuka Pemerintah Provinsi Dki Jakarta: Topik Ekonomi Dan Keuangan Daerah," *J. Sist. Inf.*, vol. 12, pp. 15–29, 2016.
- [18] B. Fry, *Visualizing data*, vol. 1. 2008.
- [19] R. Schutt and C. O'Neil, *Doing Data Science*. 2013.
- [20] S. M. Ali, N. Gupta, G. K. Nayak, and R. K. Lenka, "Big data visualization: Tools and challenges," in *2nd International Conference on Contemporary Computing and Informatics (IC3I)*, 2016, pp. 656–660.
- [21] R. Agrawal, A. Kadadi, X. Dai, and F. Andres, "Challenges and opportunities with big data visualization," in *Proceedings of the 7th International Conference on Management of computational and collective intelligence in Digital EcoSystems (MEDES '15)*, 2015, pp. 169–173.
- [22] Lo, L. Yu-Ho, Y. Ming, and H. Qu, "Learning vis tools: teaching data visualization tutorials," in *2019 IEEE Visualization Conference (VIS)*, 2019, pp. 11–15.
- [23] H. Zare and S. Emadi, "Determination of Customer Satisfaction using Improved K-means algorithm," *Soft Comput.*, vol. 24, no. 22, pp. 16947–16965, Nov. 2020.
- [24] E. Patel and D. S. Kushwaha, "Clustering Cloud Workloads: K-Means vs Gaussian Mixture Model," in *Procedia Computer Science*, 2020, vol. 171, pp. 158–167.
- [25] L. Bai, X. Cheng, J. Liang, H. Shen, and Y. Guo, "Fast density clustering strategies based on the k-means algorithm," *Pattern Recognit.*, vol. 71, pp. 375–386, Nov. 2017.
- [26] Windarto and A. Perdana, "Implementation of data mining on rice imports by major country of origin using algorithm using k-means clustering method," *Int. J. Artif. Intell. Res.*, vol. 1, no. 2, pp. 26–33, 2017.
- [27] J. Huang, Z. L. Yu, and Z. Gu, "A clustering method based on extreme learning machine," *Neurocomputing*, vol. 277, pp. 108–119, Feb. 2018.
- [28] Y. Meng, J. Liang, F. Cao, and Y. He, "A new distance with derivative information for functional k-means clustering algorithm," *Inf. Sci. (Nij.)*, vol. 463–464, pp. 166–185, Oct. 2018.